
CS 661: Knowledge Representation and Reasoning

Course code : CS 661

Course Name : Knowledge Representation and Reasoning

L-T-P-C : 3-0-0-3

Prerequisites : CS202-Data Structures & Algorithms, CS208-Mathematical Foundations of Computer Science or Equivalent or COT

Intended for : UG/PG

Elective/Core : Discipline Elective for BTech CSE, Free Elective for others

Approval: 12th Senate

Course Contents

- **Module I:** Introduction, Propositional Logic, Proof Systems. [2 Lectures]
- **Module II:** Tableau Method, Resolution Method, First Order Logic (FOL). [3 Lectures]
- **Module III:** FOL Semantics, Unification, Forward Chaining with the Rete Algorithm. [4 Lectures]
- **Module IV:** Rete example, Reification, Event Calculus, Conceptual Dependency (CD) Theory. [4 Lectures]
- **Module V:** Conceptual Dependency Theory, Mapping from Natural Language, Goal Trees. [4 Lectures]
- **Module VI:** Logic Programming with Prolog, Resolution Refutation in FOL. [4 Lectures]
- **Module VII:** SLD Resolution, Frames, Scripts. [4 Lectures]
- **Module VIII:** Goals and Plans, Description Logic (DL), Structure Matching. [4 Lectures]
- **Module IX:** Classification in DL, A-Box Reasoning, Extensions of DL, ALC. [4 Lectures]
- **Module X:** Inheritance Networks, Default Reasoning, Circumscription. [4 Lectures]
- **Module XI:** Event Calculus, Default Logic, Auto-epistemic Logic. [3 Lectures]
- **Module XII:** Epistemic Reasoning. [2 Lectures]

Textbooks:

1. Ronald J. Brachman, Hector J. Levesque, **Knowledge Representation and Reasoning**, Morgan Kaufmann, 2004.
2. Deepak Khemani, **A First Course in Artificial Intelligence**, McGraw Hill Education (India), 2013.

References:

1. Schank, Roger C., Robert P. Abelson, **Scripts, Plans, Goals, and Understanding: An Inquiry into Human Knowledge Structures**, Lawrence Erlbaum, 1977.
2. R. C. Schank and C. K. Riesbeck, **Inside Computer Understanding: Five Programs Plus Miniatures**, Lawrence Erlbaum, 1981.

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3. Murray Shanahan, **A Circumscriptive Calculus of Events**, Artif. Intell. 77(2), pp. 249-284, 1995.
 4. John F. Sowa, **Conceptual Structures: Information Processing in Mind and Machine**, Addison Wesley Publishing Company, 1984.
 5. John F. Sowa, **Knowledge Representation: Logical, Philosophical, and Computational Foundations**, Thomson Learning, 2000.
 6. Ronald Fagin, Joseph Y. Halpern, Yoram Moses, Moshe Y. Vardi, **Reasoning About Knowledge**, MIT Press, 1995.